

GRAHAM, TX, USA

WMO: 720318

Lat: 33.110N

Lon: 98.555W

Elev: 1123

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53965

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	22.8	2.9	6.6	33.4	9.5	9.3	37.0	24.7	45.9	22.0	44.1	4.6	350	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	102.4	71.8	99.9	71.4	97.5	72.0	77.4	90.3	76.5	89.3	75.5	88.4	7.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	135.1	81.2	73.2	128.8	80.4	72.4	125.0	80.2	41.7	90.4	40.6	89.3	39.8	88.4	96.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	18.0	15.9	10.7	107.2	4.2	4.3	7.7	110.3	5.2	112.8	2.8	115.2	-0.3	118.3
			10.1	80.3	3.9	5.6	7.3	84.3	5.0	87.6	2.8	90.7	0.0	94.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	44.0	48.9	57.5	65.0	72.4	81.4	84.5	84.2	77.5	65.5	54.7	45.3
	DBStd	16.77	10.02	12.08	10.41	8.66	7.79	5.01	4.47	5.09	6.56	9.49	9.83	10.07
	HDD50	737	240	156	47	7	0	0	0	0	0	9	64	214
	HDD65	2579	652	464	273	105	28	0	0	0	3	112	328	614
	CDD50	6268	54	123	281	456	694	941	1069	1062	826	489	205	68
	CDD65	2635	1	12	41	105	256	491	604	597	379	127	19	3
	CDH74	31714	37	173	467	1163	2711	5878	7848	7629	4206	1355	204	43
	CDH80	16490	4	43	142	415	1184	3115	4535	4403	2101	492	50	6
(14) Wind	WSAvg	6.3	6.2	7.0	7.9	8.2	7.0	6.3	5.3	5.0	5.0	5.8	6.1	6.1
(15) Precipitation	PrecAvg	31.6	1.4	1.9	2.5	2.8	3.8	4.1	2.8	2.3	2.8	3.1	2.1	1.6
	PrecMax	44.1	3.3	5.3	6.0	8.2	8.1	10.5	6.5	8.1	6.1	7.6	8.4	6.0
	PrecMin	20.2	0.0	0.1	0.1	0.1	0.8	0.4	0.1	0.0	0.3	0.4	0.0	0.0
	PrecStd	6.1	1.0	1.6	1.6	2.0	1.8	2.4	1.9	1.9	1.7	2.0	2.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.8	85.7	89.7	95.1	99.2	103.1	106.5	105.8	100.4	92.7	86.2	79.7
		MCWB	56.2	59.3	62.3	64.8	72.4	71.6	71.1	70.8	71.3	70.3	64.1	55.7
	2%	DB	72.3	79.3	84.0	89.2	94.7	99.3	102.2	102.4	97.1	89.7	79.4	72.4
		MCWB	53.5	59.4	61.5	64.4	69.5	70.8	71.5	72.1	71.2	69.4	61.2	54.7
	5%	DB	67.0	73.8	79.3	84.7	90.5	96.7	99.4	100.0	94.3	85.9	74.9	67.7
		MCWB	52.6	57.6	60.5	64.5	69.4	71.8	71.2	71.9	71.4	67.4	60.4	54.2
	10%	DB	62.6	68.8	74.6	80.9	87.0	93.5	97.0	97.3	91.1	81.9	71.1	62.7
		MCWB	50.5	55.4	58.6	63.2	68.9	73.0	72.7	72.6	70.7	65.3	58.1	51.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	66.3	68.1	72.3	76.8	79.4	79.0	77.8	77.1	75.6	69.3	65.2
		MCDB	68.5	75.2	80.1	82.5	87.8	92.2	92.1	90.4	88.3	84.8	75.2	70.3
	2%	WB	58.3	63.6	65.9	70.0	74.7	77.3	77.4	76.7	75.5	73.2	65.4	60.1
		MCDB	66.1	72.0	76.8	79.8	85.8	89.0	91.2	90.7	87.0	83.4	71.2	66.6
	5%	WB	54.4	60.6	63.9	68.5	73.2	76.0	76.3	75.8	74.6	71.2	63.5	56.3
		MCDB	64.9	70.3	73.0	78.2	84.3	87.7	90.0	89.8	85.8	81.2	70.4	63.7
	10%	WB	50.8	57.5	61.8	66.8	71.8	74.8	75.3	74.9	73.4	69.3	61.2	52.3
		MCDB	61.1	66.5	71.2	76.1	82.5	86.7	88.7	88.8	84.4	78.0	69.4	61.8
(35) Mean Daily Temperature Range	5% DB	MDBR	24.7	24.4	25.0	25.4	22.9	22.0	23.2	23.4	23.3	24.9	24.9	23.4
		MCDBR	32.8	31.5	29.7	30.2	27.9	25.3	26.4	26.2	26.2	27.2	28.6	31.2
	5% WB	MCWBR	18.1	16.0	13.7	13.0	9.7	6.7	6.3	6.3	7.4	11.5	14.6	16.8
		MCWBR	27.7	27.5	23.2	22.8	21.2	21.0	21.9	22.9	20.8	22.1	21.9	23.9
(40) Clear-Sky Solar Irradiance	taub		0.297	0.307	0.337	0.373	0.410	0.420	0.424	0.416	0.386	0.350	0.320	0.301
		taud	2.533	2.515	2.445	2.366	2.331	2.335	2.371	2.355	2.401	2.487	2.506	2.525
	Ebn at Noon		294	302	298	290	279	275	273	274	279	282	283	286
		Edh at Noon	26	30	34	39	41	40	39	39	36	30	27	25
(44) All-Sky Solar Radiation	RadAvg		966	1155	1530	1850	1971	2187	2188	1992	1667	1311	1020	843
	RadStd		101	173	101	150	199	187	139	124	139	195	130	102

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page